



DEVELOPERS LEVEL · SEMESTER 4

Budget Buddy

Student Workbook

Lists hold many values; loops process them — the math behind saving money.

Name: _____

THEORY

Lists & Money Math

An **array** holds many values — like a list of expenses. A **loop** repeats a step for every item, so code can total them. Money math: **income – expenses = balance**. Good budgeting means telling **needs** from **wants** and planning to save (Ontario Strand F, financial literacy).

Discuss: what expenses would a student budget track, and how would you total them?

Trace it

Read the code and give the total.

```
let costs = [4, 2, 6];
```

```
let total = 0;
```

```
for (let c of costs) { total = total + c; }
```

What is total after the loop? _____

Code we use:

array []

for (loop)

+ -

let (variable)

if

console.log

Words we are learning

- **Array** — a list of values, like each expense.
- **Loop** — repeats a step for every item (a for loop).
- **Income / Expense** — money in / money out.
- **Balance** — income minus expenses.
- **Financial literacy** — needs vs wants, saving, budgeting.

Name: _____ Date: _____

Arrays Hold Lists

An **array** is a list. Read it and answer.

```
let expenses = [5, 3, 8, 2];
```

How many items are in the array? ____ What is the first item? ____

Write an array of three expenses you might have:

```
let mine = [ ____, ____, ____ ];
```

Name: _____ Date: _____

Loops Add Them Up

A **loop** repeats for each item. Total this list by hand as the loop would.

```
let costs = [5, 3, 8, 2]; // loop adds each
```

Running total: 5 → ____ → ____ → ____ (final total)

Why is a loop better than adding by hand for 100 items?

Name: _____ Date: _____

Budget Math

Balance = **income** – **expenses**. Compute and sort needs vs wants.

Income = 20, Expenses = 14. Balance = ____

Bus fare to school	need / want
A new game	need / want
Lunch	need / want

What could you do with a positive balance?

Name: _____ **Date:** _____

Plan My Budget Buddy

Plan the tracker: what it stores, how it totals, what it tells the user.

Expenses array:

```
let expenses = [ _____ ];
```

How I total them (loop):

```
for ( _____ ) { total = total + _____; }
```

Message about their balance:

Name: _____ Date: _____

PROJECT

Budget Buddy — Plan

Build a savings/spending tracker: store expenses in an array, total them with a loop, and show the balance with a helpful message.

1. What will my tracker help with?

2. Expenses (array):

```
let expenses = [ _____ ];
```

3. Total + balance:

```
balance = income - total;
```

4. A saving tip my tool gives:
